

Scanner

The Scanner class in Java is used to read input from the user during program execution. It can read different types of data, such as integers, floating-point numbers, characters (through string input), and strings.

```
import java.util.Scanner;
```

```
Scanner sc = new Scanner(System.in);
```

- Scanner – Class name.
- sc – Object reference variable.
- new Scanner() – Creates a Scanner object.
- System.in – Reads input from the keyboard.

Important Scanner Methods

Method	Purpose	Example Input	Output
nextInt()	Reads an integer	25	25
nextFloat()	Reads a float	3.14	3.14
nextDouble()	Reads a double	45.678	45.678
nextLong()	Reads a long integer	9876543210	9876543210
nextShort()	Reads a short integer	120	120
nextByte()	Reads a byte	100	100
nextBoolean()	Reads a boolean	true	true
next()	Reads one word/token	Ashok Rajan	Ashok
nextLine()	Reads an entire line	Ashok Rajan	Ashok Rajan
next().charAt(0)	Reads the first character of a token	Ashok	A

Reading an Integer

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age = sc.nextInt();
        System.out.println("Your age is " + age);
        sc.close();
    }
}
```

Enter your age: 25
Your age is 25

Reading Different Data Types

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your name: ");
        String name = sc.next();
    }
}
```

```

System.out.print("Enter your age: ");
int age = sc.nextInt();

System.out.print("Enter your CGPA: ");
double cgpa = sc.nextDouble();

System.out.print("Are you a student? ");
boolean student = sc.nextBoolean();

System.out.println("\nStudent Details");
System.out.println("Name: " + name);
System.out.println("Age: " + age);
System.out.println("CGPA: " + cgpa);
System.out.println("Student: " + student);

sc.close();
}
}
Enter your name: Arun
Enter your age: 20
Enter your CGPA: 8.75
Are you a student? true

Student Details
Name: Arun
Age: 20
CGPA: 8.75
Student: true

```

The nextInt() and nextLine() Problem

```

import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        System.out.println("Age: " + age);
        System.out.println("Name: " + name);
        sc.close();
    }
}
Enter your age: 20
Enter your name: Age: 20
Name:

```

```

import java.util.Scanner;
class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your age: ");
        int age = sc.nextInt();

        sc.nextLine(); // Consume the leftover newline

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        System.out.println("Age: " + age);
        System.out.println("Name: " + name);

        sc.close();
    }
}

```

Enter your age: 20
 Enter your name: Ashok Rajan
 Age: 20
 Name: Ashok Rajan

Reading a Character

Java Scanner does not have a direct **nextChar()** method

```

import java.util.Scanner;

class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        System.out.println("Character: " + ch);

        sc.close();
    }
}

```

Enter a character: Ashok
 Character: A

Reading Multiple Integers Using a Loop

```

import java.util.Scanner;

class Main {
    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);

int sum = 0;

System.out.print("Enter the number of elements: ");
int n = sc.nextInt();

for (int i = 1; i <= n; i++) {
    System.out.print("Enter number " + i + ": ");
    int num = sc.nextInt();

    sum += num;
}

System.out.println("Sum = " + sum);

sc.close();
}
```

Enter the number of elements: 3
Enter number 1: 10
Enter number 2: 20
Enter number 3: 30
Sum = 60

Additional Scanner Methods

Method	Purpose
hasNextInt()	Checks whether the next input is an integer
hasNextDouble()	Checks whether the next input is a double
hasNextBoolean()	Checks whether the next input is a boolean
hasNext()	Checks whether another token is available
hasNextLine()	Checks whether another line is available
close()	Closes the Scanner object

Using hasNextInt()

```
import java.util.Scanner;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter an integer: ");

        if (sc.hasNextInt()) {
            int n = sc.nextInt();
            System.out.println("Valid integer: " + n);
        } else {
            System.out.println("Invalid input!");
        }
        sc.close();
    }
}
```

}
